

101 C PROGRAMS - COMPLETE COLLECTION

1. CALCULATOR - SUM SUB MUL DIV

```
#include <stdio.h>
int main() {
    float a, b;
    printf("Enter two numbers: ");
    scanf("%f %f", &a, &b);
    printf("Sum                = %.2f\n", a + b);
    printf("Subtraction       = %.2f\n", a - b);
    printf("Multiplication    = %.2f\n", a * b);
    if (b != 0)
        printf("Division        = %.2f\n", a / b);
    else
        printf("Division        = Infinity\n");
    return 0;
}
```

2a. LARGEST AMONG 3 NUMBERS

```
#include <stdio.h>
int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    if (a > b && a > c) printf("%d is the largest.\n", a);
    else if (b > c) printf("%d is the largest.\n", b);
    else printf("%d is the largest.\n", c);
    return 0;
}
```

2b. SMALLEST AMONG 3 NUMBERS

```
#include <stdio.h>
int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    if (a < b && a < c) printf("%d is the smallest.\n", a);
    else if (b < c) printf("%d is the smallest.\n", b);
    else printf("%d is the smallest.\n", c);
    return 0;
}
```

2c. LARGEST & SMALLEST (BOTH)

```
#include <stdio.h>
int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    if (a > b && a > c) printf("%d is the largest.\n", a);
    else if (b > c) printf("%d is the largest.\n", b);
    else printf("%d is the largest.\n", c);
    if (a < b && a < c) printf("%d is the smallest.\n", a);
    else if (b < c) printf("%d is the smallest.\n", b);
    else printf("%d is the smallest.\n", c);
    return 0;
}
```

3. AVERAGE OF 3 NUMBERS

```
#include <stdio.h>
int main() {
    float a, b, c, avg;
    printf("Enter three numbers: ");
    scanf("%f %f %f", &a, &b, &c);
    avg = (a + b + c) / 3;
    printf("Average = %.2f\n", avg);
    return 0;
}
```

4a. RIGHT ANGLE TRIANGLE AREA

```
#include <stdio.h>
int main() {
    float base, height, area;
    printf("Enter base and height: ");
    scanf("%f %f", &base, &height);
    area = (base * height) / 2;
    printf("Right Angle Triangle Area = %.2f\n", area);
    return 0;
}
```

4b. EQUILATERAL TRIANGLE AREA

```
#include <stdio.h>
int main() {
    float a, area;
    printf("Enter side length: ");
    scanf("%f", &a);
    area = (1.7320508f / 4) * a * a;
    printf("Equilateral Triangle Area = %.2f\n", area);
    return 0;
}
```

4c. SCALENE TRIANGLE AREA

```
#include <stdio.h>
int main() {
    float a, b, c, s, x, area;
    printf("Enter sides a, b, c: ");
    scanf("%f %f %f", &a, &b, &c);
    s = (a + b + c) / 2;
    x = s * (s - a) * (s - b) * (s - c);
    float guess = x / 2;
    for (int i = 0; i < 20; i++) guess = (guess + x / guess) / 2;
    area = guess;
    printf("Scalene Triangle Area = %.2f\n", area);
    return 0;
}
```

5. CIRCLE AREA & CIRCUMFERENCE

```
#include <stdio.h>
int main() {
    float r, pi = 3.1416f, area, circ;
    printf("Enter radius: ");
    scanf("%f", &r);
    area = pi * r * r;
    circ = 2 * pi * r;
    printf("Area = %.4f Sq M\n", area);
    printf("Circumference = %.4f M\n", circ);
    return 0;
}
```

6. POSITIVE OR NEGATIVE NUMBER

```
#include <stdio.h>
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n > 0) printf("%d is Positive.\n", n);
    else if (n < 0) printf("%d is Negative.\n", n);
    else printf("The number is Zero.\n");
    return 0;
}
```

7. ODD OR EVEN

```
#include <stdio.h>
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n % 2 == 0) printf("%d is Even.\n", n);
    else printf("%d is Odd.\n", n);
    return 0;
}
```

8. LEAP YEAR

```
#include <stdio.h>
int main() {
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);
    if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
        printf("%d is a Leap Year.\n", year);
    else
        printf("%d is not a Leap Year.\n", year);
    return 0;
}
```

9a. CELSIUS TO FAHRENHEIT

```
#include <stdio.h>
int main() {
    float c, f;
    printf("Enter temperature in Celsius: ");
    scanf("%f", &c);
    f = (c * 9 / 5) + 32;
    printf("%.2f C = %.2f F\n", c, f);
    return 0;
}
```

9b. FAHRENHEIT TO CELSIUS

```
#include <stdio.h>
int main() {
    float f, c;
    printf("Enter temperature in Fahrenheit: ");
    scanf("%f", &f);
    c = (f - 32) * 5 / 9;
    printf("%.2f F = %.2f C\n", f, c);
    return 0;
}
```

10a. UPPERCASE OR LOWERCASE (RANGE)

```
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a letter: ");
    scanf(" %c", &ch);
    if (ch >= 'a' && ch <= 'z') printf("%c is Lowercase.\n", ch);
    else if (ch >= 'A' && ch <= 'Z') printf("%c is Uppercase.\n", ch);
    else printf("%c is not a letter.\n", ch);
    return 0;
}
```

10b. UPPERCASE OR LOWERCASE (ASCII)

```
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a letter: ");
    scanf(" %c", &ch);
    if (ch >= 65 && ch <= 90) printf("%c is Uppercase. ASCII = %d\n", ch, ch);
    else if (ch >= 97 && ch <= 122) printf("%c is Lowercase. ASCII = %d\n", ch, ch);
    else printf("%c is not a letter.\n", ch);
    return 0;
}
```

11a. VOWEL OR CONSONANT - IF

```
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a character: ");
    scanf(" %c", &ch);
    if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' || ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U')
        printf("%c is a Vowel.\n", ch);
    else
        printf("%c is a Consonant.\n", ch);
    return 0;
}
```

11b. VOWEL OR CONSONANT - SWITCH

```
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a character: ");
    scanf(" %c", &ch);
    switch (ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            printf("%c is a Vowel.\n", ch);
            break;
        default:
            printf("%c is a Consonant.\n", ch);
    }
    return 0;
}
```

12a. PRINT 10 TIMES - FOR LOOP

```
#include <stdio.h>
int main() {
    for (int i = 1; i <= 10; i++) printf("Bangladesh\n");
    return 0;
}
```

12b. PRINT 10 TIMES - WHILE LOOP

```
#include <stdio.h>
int main() {
    int i = 1;
    while (i <= 10) { printf("Bangladesh\n"); i++; }
    return 0;
}
```

12c. PRINT 10 TIMES - DO-WHILE

```
#include <stdio.h>
int main() {
    int i = 1;
    do { printf("Bangladesh\n"); i++; } while (i <= 10);
    return 0;
}
```

12d. PRINT TEXT N TIMES - FOR

```
#include <stdio.h>
int main() {
    char text[100];
    int n;
    printf("Enter text: ");
    scanf("%[^\n]", text);
    printf("How many times? ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) printf("%s\n", text);
    return 0;
}
```

12e. PRINT TEXT N TIMES - WHILE

```
#include <stdio.h>
int main() {
    char text[100];
    int n, i = 1;
    printf("Enter text: ");
    scanf("%[^\n]", text);
    printf("How many times? ");
    scanf("%d", &n);
    while (i <= n) { printf("%s\n", text); i++; }
    return 0;
}
```

12f. PRINT TEXT N TIMES - DO-WHILE

```
#include <stdio.h>
int main() {
    char text[100];
    int n, i = 1;
    printf("Enter text: ");
    scanf(" %[^\n]", text);
    printf("How many times? ");
    scanf("%d", &n);
    do { printf("%s\n", text); i++; } while (i <= n);
    return 0;
}
```

13a. PRINT 1 TO 10

```
#include <stdio.h>
int main() {
    for (int i = 1; i <= 10; i++) printf("%d ", i);
    printf("\n");
    return 0;
}
```

13b. PRINT 1 TO N

```
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) printf("%d ", i);
    printf("\n");
    return 0;
}
```

14a. SUM OF 1 TO 10

```
#include <stdio.h>
int main() {
    int sum = 0;
    for (int i = 1; i <= 10; i++) sum += i;
    printf("Sum = %d\n", sum);
    return 0;
}
```

14b. SUM OF 1 TO N

```
#include <stdio.h>
int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) sum += i;
    printf("Sum = %d\n", sum);
    return 0;
}
```

15. PRIME NUMBER CHECK

```
#include <stdio.h>
int main() {
    int n, isPrime = 1;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n < 2) isPrime = 0;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) { isPrime = 0; break; }
    }
    if (isPrime) printf("%d is a Prime number.\n", n);
    else printf("%d is not a Prime number.\n", n);
    return 0;
}
```

16. PERFECT NUMBER CHECK

```
#include <stdio.h>
int main() {
    int n, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    for (int i = 1; i < n; i++)
        if (n % i == 0) sum += i;
    if (sum == n) printf("%d is a Perfect number.\n", n);
    else printf("%d is not a Perfect number.\n", n);
    return 0;
}
```

17. SUM OF DIGITS

```
#include <stdio.h>
int main() {
    int n, sum = 0, temp;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while (temp != 0) {
        sum += temp % 10;
        temp /= 10;
    }
    printf("Sum of digits = %d\n", sum);
    return 0;
}
```


18. REVERSE A NUMBER

```
#include <stdio.h>
int main() {
    int n, rev = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    while (n != 0) {
        rev = rev * 10 + n % 10;
        n /= 10;
    }
    printf("Reversed number = %d\n", rev);
    return 0;
}
```

19. PALINDROME NUMBER

```
#include <stdio.h>
int main() {
    int n, rev = 0, temp;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while (temp != 0) {
        rev = rev * 10 + temp % 10;
        temp /= 10;
    }
    if (rev == n) printf("%d is a Palindrome.\n", n);
    else printf("%d is not a Palindrome.\n", n);
    return 0;
}
```

20. ARMSTRONG NUMBER

```
#include <stdio.h>
int main() {
    int n, temp, digits = 0, sum = 0, t;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while (temp != 0) { digits++; temp /= 10; }
    temp = n;
    while (temp != 0) {
        t = temp % 10;
        int p = 1;
        for (int i = 0; i < digits; i++) p *= t;
        sum += p;
        temp /= 10;
    }
    if (sum == n) printf("%d is Armstrong.\n", n);
    else printf("%d is not Armstrong.\n", n);
    return 0;
}
```

21. STRONG NUMBER

```
#include <stdio.h>
int main() {
    int n, temp, sum = 0, digit, fact;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while (temp != 0) {
        digit = temp % 10;
        fact = 1;
        for (int i = 1; i <= digit; i++) fact *= i;
        sum += fact;
        temp /= 10;
    }
    if (sum == n) printf("%d is Strong.\n", n);
    else printf("%d is not Strong.\n", n);
    return 0;
}
```

22. GCD AND LCM

```
#include <stdio.h>
int main() {
    int a, b, x, y, gcd, lcm;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    x = a; y = b;
    while (y != 0) { int t = y; y = x % y; x = t; }
    gcd = x;
    lcm = (a * b) / gcd;
    printf("GCD = %d\nLCM = %d\n", gcd, lcm);
    return 0;
}
```

23. FACTORIAL

```
#include <stdio.h>
int main() {
    int n;
    long long fact = 1;
    printf("Enter a number: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) fact *= i;
    printf("%d! = %lld\n", n, fact);
    return 0;
}
```

24. SERIES: $1! + 2! + \dots + n!$

```
#include <stdio.h>
int main() {
    int n;
    long long sum = 0, fact = 1;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        fact *= i;
        sum += fact;
    }
    printf("Sum = %lld\n", sum);
    return 0;
}
```

25. SERIES: $2! + 4! + \dots + n!$

```
#include <stdio.h>
int main() {
    int n;
    long long sum = 0, fact;
    printf("Enter n (even): ");
    scanf("%d", &n);
    for (int i = 2; i <= n; i += 2) {
        fact = 1;
        for (int j = 1; j <= i; j++) fact *= j;
        sum += fact;
    }
    printf("Sum = %lld\n", sum);
    return 0;
}
```

26. SERIES: $1! + 3! + \dots + n!$

```
#include <stdio.h>
int main() {
    int n;
    long long sum = 0, fact;
    printf("Enter n (odd): ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i += 2) {
        fact = 1;
        for (int j = 1; j <= i; j++) fact *= j;
        sum += fact;
    }
    printf("Sum = %lld\n", sum);
    return 0;
}
```

27a. FIBONACCI - PRINT TERMS

```
#include <stdio.h>
int main() {
    int n, a = 0, b = 1, c;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    printf("Series: %d %d", a, b);
    for (int i = 3; i <= n; i++) {
        c = a + b; printf(" %d", c); a = b; b = c;
    }
    printf("\n");
    return 0;
}
```

27b. FIBONACCI - PRINT & SUM

```
#include <stdio.h>
int main() {
    int n, a = 0, b = 1, c;
    long long sum = a + b;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    printf("Series: %d %d", a, b);
    for (int i = 3; i <= n; i++) {
        c = a + b; printf(" %d", c);
        sum += c; a = b; b = c;
    }
    printf("\nSum = %lld\n", sum);
    return 0;
}
```

28. SERIES: 1 + 2 + ... + n

```
#include <stdio.h>
int main() {
    int n; long long sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) sum += i;
    printf("Sum = %lld\n", sum);
    return 0;
}
```

29. SERIES: 1 + 3 + 5 + ... + n

```
#include <stdio.h>
int main() {
    int n; long long sum = 0;
    printf("Enter n (odd): ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i += 2) sum += i;
    printf("Sum = %lld\n", sum);
    return 0;
}
```

30. SERIES: $2 + 4 + 6 + \dots + n$

```
#include <stdio.h>
int main() {
    int n; long long sum = 0;
    printf("Enter n (even): ");
    scanf("%d", &n);
    for (int i = 2; i <= n; i += 2) sum += i;
    printf("Sum = %lld\n", sum);
    return 0;
}
```

31. SERIES: $1^1 + 2^2 + \dots + n^n$

```
#include <stdio.h>
int main() {
    int n; long long sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        long long p = 1;
        for (int j = 0; j < i; j++) p *= i;
        sum += p;
    }
    printf("Sum = %lld\n", sum);
    return 0;
}
```

32. SERIES: $1^2 + 2^2 + \dots + n^2$

```
#include <stdio.h>
int main() {
    int n; long long sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) sum += (long long)i * i;
    printf("Sum = %lld\n", sum);
    return 0;
}
```

33. SERIES: $1^2 + 3^2 + \dots + n^2$

```
#include <stdio.h>
int main() {
    int n; long long sum = 0;
    printf("Enter n (odd): ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i += 2) sum += (long long)i * i;
    printf("Sum = %lld\n", sum);
    return 0;
}
```

34. SERIES: $2^2 + 4^2 + \dots + n^2$

```
#include <stdio.h>
int main() {
    int n; long long sum = 0;
    printf("Enter n (even): ");
    scanf("%d", &n);
    for (int i = 2; i <= n; i += 2) sum += (long long)i * i;
    printf("Sum = %lld\n", sum);
    return 0;
}
```

35. SERIES: $1 + 2 + 4 + 8 + \dots$

```
#include <stdio.h>
int main() {
    int n; long long sum = 0, term = 1;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) { sum += term; term *= 2; }
    printf("Sum = %lld\n", sum);
    return 0;
}
```

36. SERIES: $1 + 3 + 9 + 27 + \dots$

```
#include <stdio.h>
int main() {
    int n; long long sum = 0, term = 1;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) { sum += term; term *= 3; }
    printf("Sum = %lld\n", sum);
    return 0;
}
```

37. SERIES: $1/1 + 1/2 + \dots + 1/n$

```
#include <stdio.h>
int main() {
    int n; float sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) sum += 1.0f / i;
    printf("Sum = %.4f\n", sum);
    return 0;
}
```

38. SERIES: $1/1^2 + 1/2^2 + \dots$

```
#include <stdio.h>
int main() {
    int n; float sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) sum += 1.0f / ((float)i * i);
    printf("Sum = %.4f\n", sum);
    return 0;
}
```

39. PATTERN - STARS INCREASING

```
*
**
***
****
*****
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("*");
        printf("\n");
    }
    return 0;
}
```

40. PATTERN - NUMBERS INCREASING

```
1
12
123
1234
12345
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%d", c);
        printf("\n");
    }
    return 0;
}
```

41. PATTERN - REPEATED ROW NUM

1
22
333
4444
55555

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%d", r);
        printf("\n");
    }
    return 0;
}
```

42. PATTERN - 1 0 ALT START 1

1
10
101
1010
10101

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%d", (c % 2 == 1) ? 1 : 0);
        printf("\n");
    }
    return 0;
}
```

43. PATTERN - 1 0 ALT ODD=1 EVEN=0

1
00
111
0000
11111

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        int v = (r % 2 == 1) ? 1 : 0;
        for (int c = 1; c <= r; c++) printf("%d", v);
        printf("\n");
    }
    return 0;
}
```


44. PATTERN - REPEATED ALPHABET

A
BB
CCC
DDDD
EEEE

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%c", 'A' + r - 1);
        printf("\n");
    }
    return 0;
}
```

45. PATTERN - ALPHABETS INC

A
AB
ABC
ABCD
ABCDE

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%c", 'A' + c - 1);
        printf("\n");
    }
    return 0;
}
```

46. PATTERN - STARS DECREASING

**
*

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int c = 1; c <= r; c++) printf("*");
        printf("\n");
    }
    return 0;
}
```

47. PATTERN - NUMBERS DECREASING

```
12345
1234
123
12
1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int c = 1; c <= r; c++) printf("%d", c);
        printf("\n");
    }
    return 0;
}
```

48. PATTERN - REP ROW DECREASING

```
55555
4444
333
22
1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int c = 1; c <= r; c++) printf("%d", r);
        printf("\n");
    }
    return 0;
}
```

49. PATTERN - 1 0 ALT DECREASING

```
10101
1010
101
10
1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int c = 1; c <= r; c++) printf("%d", (c % 2 == 1) ? 1 : 0);
        printf("\n");
    }
    return 0;
}
```

50. PATTERN - ROW 1/0 DECREASING

```
11111
0000
111
00
1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        int v = (r % 2 == 1) ? 1 : 0;
        for (int c = 1; c <= r; c++) printf("%d", v);
        printf("\n");
    }
    return 0;
}
```

51. PATTERN - ALPHABETS DEC

```
ABCDE
ABCD
ABC
AB
A
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int c = 1; c <= r; c++) printf("%c", 'A' + c - 1);
        printf("\n");
    }
    return 0;
}
```

52. PATTERN - REP ALPHABET DEC

```
EEEE
DDDD
CCC
BB
A
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int c = 1; c <= r; c++) printf("%c", 'A' + r - 1);
        printf("\n");
    }
    return 0;
}
```

53. PATTERN - STARS DIAMOND

```
*  
**  
***  
**  
*
```

```
#include <stdio.h>  
int main() {  
    int n; scanf("%d", &n);  
    for (int r = 1; r <= n; r++) {  
        for (int c = 1; c <= r; c++) printf("*");  
        printf("\n");  
    }  
    for (int r = n - 1; r >= 1; r--) {  
        for (int c = 1; c <= r; c++) printf("*");  
        printf("\n");  
    }  
    return 0;  
}
```

54. PATTERN - NUMBERS DIAMOND

```
1  
12  
123  
12  
1
```

```
#include <stdio.h>  
int main() {  
    int n; scanf("%d", &n);  
    for (int r = 1; r <= n; r++) {  
        for (int c = 1; c <= r; c++) printf("%d", c);  
        printf("\n");  
    }  
    for (int r = n - 1; r >= 1; r--) {  
        for (int c = 1; c <= r; c++) printf("%d", c);  
        printf("\n");  
    }  
    return 0;  
}
```

55. PATTERN - REP ROW NUM DIAMOND

```
1
22
333
22
1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%d", r);
        printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
        for (int c = 1; c <= r; c++) printf("%d", r);
        printf("\n");
    }
    return 0;
}
```

56. PATTERN - 1/0 START 1 DIAMOND

```
1
10
101
10
1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%d", (c % 2 == 1) ? 1 : 0);
        printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
        for (int c = 1; c <= r; c++) printf("%d", (c % 2 == 1) ? 1 : 0);
        printf("\n");
    }
    return 0;
}
```

57. PATTERN - ROW 1/0 DIAMOND

```
1
00
111
00
1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        int v = (r % 2 == 1) ? 1 : 0;
        for (int c = 1; c <= r; c++) printf("%d", v);
        printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
        int v = (r % 2 == 1) ? 1 : 0;
        for (int c = 1; c <= r; c++) printf("%d", v);
        printf("\n");
    }
    return 0;
}
```

58. PATTERN - REP ALPHABET DIAMOND

```
A
BB
CCC
BB
A
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%c", 'A' + r - 1);
        printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
        for (int c = 1; c <= r; c++) printf("%c", 'A' + r - 1);
        printf("\n");
    }
    return 0;
}
```

59. PATTERN - ALPHABET SEQ DIAMOND

A
AB
ABC
AB
A

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%c", 'A' + c - 1);
        printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
        for (int c = 1; c <= r; c++) printf("%c", 'A' + c - 1);
        printf("\n");
    }
    return 0;
}
```

60. PATTERN - ALPHABET SEQ DIAMOND

/ Same as 59 */*

/ See program 59 above */*

61. PATTERN - RIGHT-ALIGNED STARS

*
**


```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("*");
        printf("\n");
    }
    return 0;
}
```

62. PATTERN - RIGHT-ALIGNED NUMBERS

```
1
12
123
1234
12345
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%d", c);
        printf("\n");
    }
    return 0;
}
```

63. PATTERN - RIGHT-ALIGNED REP ROW

```
1
22
333
4444
55555
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%d", r);
        printf("\n");
    }
    return 0;
}
```

64. PATTERN - RIGHT-ALIGNED 1/0 START 1

```
1
10
101
1010
10101
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%d", (c % 2 == 1) ? 1 : 0);
        printf("\n");
    }
    return 0;
}
```


65. PATTERN - RIGHT-ALIGNED ROW 1/0

```
1
00
111
0000
11111
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        int v = (r % 2 == 1) ? 1 : 0;
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%d", v);
        printf("\n");
    }
    return 0;
}
```

66. PATTERN - RIGHT-ALIGNED REP ALPHA

```
A
BB
CCC
DDDD
EEEE
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%c", 'A' + r - 1);
        printf("\n");
    }
    return 0;
}
```

67. PATTERN - RIGHT-ALIGNED ALPHA SEQ

```
A
AB
ABC
ABCD
ABCDE
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%c", 'A' + c - 1);
        printf("\n");
    }
    return 0;
}
```

68. PATTERN - LEFT-INDENT STARS DEC

```
****
***
**
*
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("*");
        printf("\n");
    }
    return 0;
}
```

69. PATTERN - LEFT-INDENT NUMBERS DEC

```
12345
1234
123
12
1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%d", c);
        printf("\n");
    }
    return 0;
}
```

70. PATTERN - LEFT-INDENT REP ROW DEC

```
55555
4444
333
22
1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%d", r);
        printf("\n");
    }
    return 0;
}
```

71. PATTERN - LEFT-INDENT 1/0 DEC

```
10101
 1010
   101
    10
     1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%d", (c % 2 == 1) ? 1 : 0);
        printf("\n");
    }
    return 0;
}
```

72. PATTERN - LEFT-INDENT ROW 1/0 DEC

```
11111
0000
 111
   00
    1
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        int v = (r % 2 == 1) ? 1 : 0;
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%d", v);
        printf("\n");
    }
    return 0;
}
```

73. PATTERN - LEFT-INDENT REP ALPHA DEC

```
EEEE
DDDD
CCC
BB
A
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%c", 'A' + r - 1);
        printf("\n");
    }
    return 0;
}
```

74. PATTERN - LEFT-INDENT ALPHA SEQ DEC

ABCDE
ABCD
ABC
AB
A

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= r; c++) printf("%c", 'A' + c - 1);
        printf("\n");
    }
    return 0;
}
```

75. PATTERN - SOLID STAR SQUARE


```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= n; c++) printf("*");
        printf("\n");
    }
    return 0;
}
```

76. PATTERN - NUMBER SEQUENCE SQUARE

12345
12345
12345

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= n; c++) printf("%d", c);
        printf("\n");
    }
    return 0;
}
```

77. PATTERN - ROW NUMBER SQUARE

```
1111
2222
3333
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= n; c++) printf("%d", r);
        printf("\n");
    }
    return 0;
}
```

78. PATTERN - 1010 COL ALT SQUARE

```
1010
1010
1010
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= n; c++) printf("%d", (c % 2 == 1) ? 1 : 0);
        printf("\n");
    }
    return 0;
}
```

79. PATTERN - 1111/0000 ROW ALT SQUARE

```
1111
0000
1111
```

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        int v = (r % 2 == 1) ? 1 : 0;
        for (int c = 1; c <= n; c++) printf("%d", v);
        printf("\n");
    }
    return 0;
}
```

80. PATTERN - REPEATED ALPHABET SQUARE

AAAA
BBBB
CCCC

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= n; c++) printf("%c", 'A' + r - 1);
        printf("\n");
    }
    return 0;
}
```

90. PATTERN - ALPHABET SEQ SQUARE

ABCD
ABCD
ABCD

```
#include <stdio.h>
int main() {
    int n; scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= n; c++) printf("%c", 'A' + c - 1);
        printf("\n");
    }
    return 0;
}
```

91. PATTERN - SEQUENTIAL NUMS TRIANGLE

1
2 3
4 5 6

```
#include <stdio.h>
int main() {
    int n, count = 1;
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%d ", count++);
        printf("\n");
    }
    return 0;
}
```

92. PATTERN - SEQUENTIAL FROM 0

```
0
1 2
3 4 5
```

```
#include <stdio.h>
int main() {
    int n, count = 0;
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%d ", count++);
        printf("\n");
    }
    return 0;
}
```

93. PATTERN - MULTIPLICATION TRIANGLE

```
1
2 4
4 6 9
```

```
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int c = 1; c <= r; c++) printf("%d ", r * c);
        printf("\n");
    }
    return 0;
}
```

94. PATTERN - PYRAMID OF STARS

```
*
***
*****
```

```
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
        for (int s = 1; s <= n - r; s++) printf(" ");
        for (int c = 1; c <= (2 * r - 1); c++) printf("*");
        printf("\n");
    }
    return 0;
}
```

95. FACTORIAL USING RECURSION

```
#include <stdio.h>
long long fact(int n) {
    if (n == 0 || n == 1) return 1;
    return n * fact(n - 1);
}
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    printf("%d! = %lld\n", n, fact(n));
    return 0;
}
```

96. FIBONACCI USING RECURSION

```
#include <stdio.h>
int fib(int n) {
    if (n <= 1) return n;
    return fib(n - 1) + fib(n - 2);
}
int main() {
    int n;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    printf("Series: ");
    for (int i = 0; i < n; i++) printf("%d ", fib(i));
    printf("\n");
    return 0;
}
```

97. CALCULATE POWER USING RECURSION

```
#include <stdio.h>
long long power(int base, int exp) {
    if (exp == 0) return 1;
    return base * power(base, exp - 1);
}
int main() {
    int base, exp;
    printf("Enter base and exponent: ");
    scanf("%d %d", &base, &exp);
    printf("%d^%d = %lld\n", base, exp, power(base, exp));
    return 0;
}
```


98a. ARRAY - FIND MAXIMUM

```
#include <stdio.h>
int main() {
    int n;
    printf("How many numbers? ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d numbers: ", n);
    for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
    int max = arr[0];
    for (int i = 1; i < n; i++)
        if (arr[i] > max) max = arr[i];
    printf("Maximum = %d\n", max);
    return 0;
}
```

98b. ARRAY - FIND MINIMUM

```
#include <stdio.h>
int main() {
    int n;
    printf("How many numbers? ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d numbers: ", n);
    for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
    int min = arr[0];
    for (int i = 1; i < n; i++)
        if (arr[i] < min) min = arr[i];
    printf("Minimum = %d\n", min);
    return 0;
}
```

99. ARRAY - FIND BOTH MAX AND MIN

```
#include <stdio.h>
int main() {
    int n;
    printf("How many numbers? ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d numbers: ", n);
    for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
    int max = arr[0], min = arr[0];
    for (int i = 1; i < n; i++) {
        if (arr[i] > max) max = arr[i];
        if (arr[i] < min) min = arr[i];
    }
    printf("Maximum = %d\nMinimum = %d\n", max, min);
    return 0;
}
```

100. SWAP TWO NUMBERS WITHOUT TEMP

```
#include <stdio.h>
int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    printf("Before: a = %d, b = %d\n", a, b);
    a = a + b;
    b = a - b;
    a = a - b;
    printf("After : a = %d, b = %d\n", a, b);
    return 0;
}
```

101a. CONTINUE EXAMPLE

```
#include <stdio.h>
int main() {
    for (int i = 1; i <= 10; i++) {
        if (i % 3 == 0) continue;
        printf("%d ", i);
    }
    printf("\n");
    return 0;
}
```

101b. BREAK EXAMPLE

```
#include <stdio.h>
int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 6) break;
        printf("%d ", i);
    }
    printf("\n");
    return 0;
}
```